

Soldering Bootcamp

Order, stage, set up, and confirm before the meeting — plus the MATE pre-watch homework.

One three-hour session — all hands on irons. (The kit comes out in Class 1.3.)

ORDER TODAY SeaMATE lab kits (2+ wk lead), battery, stations, safety gear.	T-1 WEEK Assign MATE pre-watch. Buy consumables. Preview the presentations.	T-1 DAY Stage the room: stations, mats, ventilation, kits at each seat.	T-0 Safety brief. Irons hot. Stations live.
---	---	--	--

ORDER AHEAD · LEAD-TIME ALARMS

- ☐ **ORDER NOW** SeaMATE Wire Soldering Lab Kit — one per student (×6). \$8 ea (\$48; 10-pack \$65). 2+ wk lead. [seamate.org](#)
- ☐ **ORDER NOW** SeaMATE PufferFish Practice Board — one per student. 10-pack \$65 (best value). Needs a 9 V battery each. 2+ wk lead.
- ☐ **ORDER NOW** 12 V SLA battery, charger, enclosure, Powerpole leads — in hand before the electronics class. (\$125)
- ☐ Two soldering stations, irons, consumables — this meeting. (\$170)
- ☐ Two multimeters + hand-tool set: strippers, cutters, pliers. (\$130)
- ☐ Heat-shrink, wire, zip ties, build consumables. (\$55)
- ☐ Safety gear — fume fan, first-aid kit, fire extinguisher, mats. (\$100)

PRE-WATCH HOMEWORK · ASSIGN BEFORE CLASS

- ☐ MATE: *Using Tools* (20–30 min)
- ☐ MATE: *Soldering & Waterproofing Wires*
- ☐ Optional refresher: *How to Solder* (22 min, YouTube)

Let MATE teach the concepts; protect meeting time for hands-on.

STAGE IN THE ROOM

- ☐ 2 soldering stations, irons, tip cleaner / sponges
- ☐ A Wire Soldering Lab Kit + Practice Board at each seat (+ a 9 V battery per board)
- ☐ Solder, plus helping-hands clamps if available
- ☐ Heat-shrink assortment + heat source
- ☐ Wire strippers, flush cutters, needle-nose pliers
- ☐ Project notebook + a phone/camera for the photo log

PREP OWNERS

- ☐ **CFO + ADVISOR** — order SeaMATE kits + long-lead items **today**
- ☐ **ADVISOR** — assign MATE pre-watch, buy consumables (T–1 wk)

SAY WATT ROBOTICS · ROV TEAM

SET UP · BEFORE KIDS ARRIVE

- ☐ Power strips at each station; cords off the floor
- ☐ Ventilation on — fume fan or open window
- ☐ Heat-safe mats under every iron; clear the benches
- ☐ Safety glasses laid out at each seat
- ☐ First-aid kit + extinguisher visible

CUE UP · IN-ROOM REFERENCE

- ☐ MATE: *Solder & Seal Connectors* and *Soldering Components to a PCB*
- ☐ Instructor Notes — esp. the open vs. short circuit grounding

STUDENTS BRING

- ☐ Watched the MATE pre-watch homework
- ☐ Closed-toe shoes (required at the bench); hair tied back
- ☐ GitHub login ready (Software Lead)

CONFIRM · GATING

- ☐ Battery + charger ordered — in hand before the electronics class
- ☐ Pool prerequisites moving — lifeguard, swim checks, host rules in writing
- ☐ DISL: confirm 2027 dates + pool depth (Rachel McDonald)

- ☐ **ADVISOR** — stage room, irons, ventilation (T–1 day)
- ☐ **CEO + ADVISOR** — run safety brief, open stations (T–0)

CLASS 1.2 — SOLDERING BOOTCAMP · PREP SHEET